

Rancher Management Server Setup

Prerequisites

- **Installation of Necessary Tools on Console Machine**

- Install Git and Tmux:

```
sudo apt update && sudo apt install git tmux -y
```

- Install Ansible:

```
sudo apt update && sudo apt install software-properties-common -y && sudo add-apt-repository  
--yes --update ppa:ansible/ansible && sudo apt install ansible -y
```

- Install Helm:

```
wget https://get.helm.sh/helm-v3.16.2-linux-amd64.tar.gz && tar -zxvf helm-v3.16.2-linux-  
amd64.tar.gz && sudo mv linux-amd64/helm /bin/
```

- Verify Installed Tools:

```
ansible --version  
helm version
```

- **Operating System:**

- Ensure that all the OS system is running with `Ubuntu 24.04.1 LTS` version:

```
PRETTY_NAME="Ubuntu 24.04.1 LTS"  
NAME="Ubuntu"  
VERSION_ID="24.04"  
VERSION="24.04.1 LTS (Noble Numbat)"  
VERSION_CODENAME=noble  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=noble  
LOGO=ubuntu-logo
```

- **Infrastructure Requirements:**

- Set up Virtual Machines (VMs) for the cluster.
- Install `nfs-common` package on all the nodes of the Kubernetes cluster.

```
sudo apt install nfs-common -y
```

- Configure an NFS volume and mount it on the primary and subsequent control plane nodes at `/mnt/rancher-k8s-data/`. Ensure that the mount point for NFS volumes is added to `/etc/fstab` to ensure automatic mounting upon system boot.

```
$ cat /etc/fstab  
....  
....  
172.16.119.11:/rancher-k8s-data /mnt/rancher-k8s-data nfs rw, sync, noac, vers=4 0 0
```

- Set up a load balancer with a domain and IP for the Kubernetes cluster.
- Expose the following ports on the load balancer.
 - 9345/tcp
 - 443/tcp
 - 6443/tcp

The load balancer route:

Load balancer host	Load balancer		Kubernetes node ports
rancher.nsis.nira.go.ug	9345/tcp	----->	<controlplane nodes> : 9345/tcp
rancher.nsis.nira.go.ug	6443/tcp	----->	<controlplane nodes> : 6443/tcp
rancher.nsis.nira.go.ug	443/tcp	----->	<kubernetes nodes> : 30443/tcp
keycloak.nsis.nira.go.ug	443/tcp	----->	<kubernetes nodes> : 30443/tcp
rancher-kibana.nsis.nira.go.ug	443/tcp	----->	<kubernetes nodes> : 30080/tcp

For rancher.nsis.nira.go.ug and keycloak.nsis.nira.go.ug , do not offload SSL on the load balancer. Ensure that the SSL certificates are forwarded with the request from the load balancer to the Kubernetes node ports on 30443 .

- Obtain valid SSL certificates and domain names:
 - Domains: rancher.nsis.nira.go.ug , keycloak.nsis.nira.go.ug
 - A wildcard SSL certificate for *.nsis.nira.go.ug .
- Disable swap on all the Kubernetes nodes.

```
sudo swapoff -a
```

Remove swap file entry from /etc/fstab file and remove the swap file /swap.img from OS.

```
/swap.img    none    swap    sw      0      0
```

Ensure to reboot the OS.

- Expose the necessary ports for Kubernetes nodes to communicate to each other.
 - Control plane node / Subsequent control plane nodes:
 - Kubernetes API: 9345/tcp
 - RKE2 supervisor API: 6443/tcp
 - ETCD client port: 2379/tcp
 - ETCD peer port: 2380/tcp
 - ETCD metrics port: 2381/tcp
 - Kubelet metrics: 10250/tcp
 - Canal CNI with VXLAN: 8472/udp
 - Canal CNI health checks: 9099/tcp
- Start a tmux session on console machine to ensure session persistence. This allows you to resume your shell in case of a network disruption between your local system and the console machine.

```
tmux new -s <session-name>
```

- Configure ssh, from the console machine to all the Kubernetes nodes.
 - Generate SSH keys on your console machine:

```
ssh-keygen -t rsa
```

- Set up passwordless SSH access by copying your SSH key to remote machines using the following steps.
 - Install `sshpass` on console machine.

```
sudo apt-get install sshpass -y
```

- Update the variables USERNAME, PASSWORD, and VM_LIST in the command below, then execute it from the console machine.

USERNAME : The username for SSH authentication.

PASSWORD : The corresponding password for the user.

VM_LIST : A space-separated list of IP addresses or hostnames of target machines, enclosed in double quotes.

```
VM_LIST=("<node-1>" "<node-2>" "<node-3>" "<node-4>")
USERNAME=<user-name>
PASSWORD=<password>
for VM in "${VM_LIST[@]}; do
echo "Server : $VM "
sshpass -p "$PASSWORD" ssh-copy-id -o StrictHostKeyChecking=no $USERNAME@$VM
done
```

```
VM_LIST=("172.16.112.11" "172.16.112.12" "172.16.112.13")
USERNAME=YYYYY
PASSWORD=XXXXX
```

Kubernetes Setup via Ansible Playbook

Steps to Deploy the RKE2 Kubernetes Cluster:

1. Clone the repository:

```
cd ~/
git clone https://github.com/tf-nira/k8s-infra.git -b develop
cd ~/k8s-infra/k8-cluster/on-prem/rke2/ansible
```

2. Create and update the inventory file:

- Create inventory hosts file from sample file

```
cp hosts.ini.sample hosts.ini
```

- Define host details in `control_plane_primary` and `control_plane_subsequent` groups in `hosts.ini` :

```
ansible_host=<internal_ip>
ansible_user=<username>
ansible_ssh_private_key_file=/home/<username>/.ssh/id_rsa
```

- Comment out the `agents` group in `hosts.ini` file.
- Update variables in `hosts.ini` file:
 - `cluster_domain` : e.g., `rancher`
 - `rke2_token` : Unique token for nodes to join the kubernetes cluster.
 - `ETCD_BACKUP_DIR` : Path for etcd snapshots, e.g., `/mnt/rancher-k8s-data` (snapshots stored in `/mnt/rancher-k8s-data/etcd/snapshots`)
 - `AUDIT_BACKUP_DIR` : Path for audit logs, e.g., `/mnt/rancher-k8s-data` (logs stored in `/mnt/rancher-k8s-data/audit`)
 - `LB_DOMAIN` : e.g., `rancher.nsis.nira.go.ug`
 - `LB_IP` : e.g., `172.16.130.71`

3. Run the following command to initiate the setup of the RKE2 Kubernetes cluster:

```
ansible-playbook -i hosts.ini main.yaml
```

Upon successful execution, the Kubernetes cluster's kubeconfig file will be available on all control plane nodes at `/etc/rancher/rke2/rke2.yaml`

4. To manage the cluster from a console machine:

- Copy the kubeconfig file from a control plane node to the console machine.
- Set read only permission to the kubeconfig file.

```
chmod 400 <kube-config-file>
```

- Update the server URL within the copied kubeconfig file to point to the load balancers domain:

```
server: https://<load-balancer-domain>:6443
```

eg: replace `127.0.0.1` to `load-balancer-domain`

```
apiVersion: v1
clusters:
- cluster:
    ....
    ....
    server: https://rancher.nsis.nira.go.ug:6443
    ....
```

5. Once the playbook executes successfully, copy the `kubectl` binary from one of the control plane nodes to the console machine. Update username and control plane host in the below command to copy the kubectl executable file to console machine.

```
scp <username>@<control-plane-host>:/var/lib/rancher/rke2/bin/kubectl ./kubectl
```

Move the `kubectl` executable file to `/bin/` directory.

```
sudo chmod +x ./kubectl && sudo mv ./kubectl /bin/
```

Verify the kubectl version. i.e., `v1.28.9+rke2r1`

```
kubectl version
```

6. Point kubectl to newly created cluster config file.

```
export KUBECONFIG=/path/to/kubeconfig/file
```

eg.

```
export KUBECONFIG=/home/ubuntu/k8s-infra/k8-cluster/on-prem/rke2/ansible/playbook/kubeconfigs/rancher-control-plane-1.yaml`
```

Access the kubernetes cluster via the below command:

```
kubectl get nodes
```

Note: By default, the RKE2 server and agent services are disabled to prevent additional system load during OS boot.

Setup Ingress as NodePort

- Navigate to ingress-nginx directory

```
cd ~/k8s-infra/ingress/ingress-nginx
```

- Add helm repo and update the same:

```
helm repo add ingress-nginx https://kubernetes.github.io/ingress-nginx
helm repo update
```

- Helm install ingress-nginx

```
helm install ingress-nginx ingress-nginx/ingress-nginx --namespace ingress-nginx --version 4.0.18
--create-namespace -f ingress-nginx-np.values.yaml
```

```
kubectl get all -n ingress-nginx
```

Install nfs-csi

- Navigate to nfs directory.

```
cd ~/k8s-infra/storage-class/nfs
```

- Run `./install-nfs-csi.sh` to deploy NFS client provisioner. Ensure to provide valid NFS server and NFS server path.

```
./install-nfs-csi.sh
.....
Please provide NFS SERVER: <NFS-SERVER>
Please provide NFS Path: <NFS-SERVER-PATH>
```

Rancher deployment

- Navigate to rancher-ui directory.

```
cd ~/k8s-infra/apps/rancher-ui/
```

- Install rancher using Helm, update `hostname` in `rancher-values.yaml` and run the following command to install.

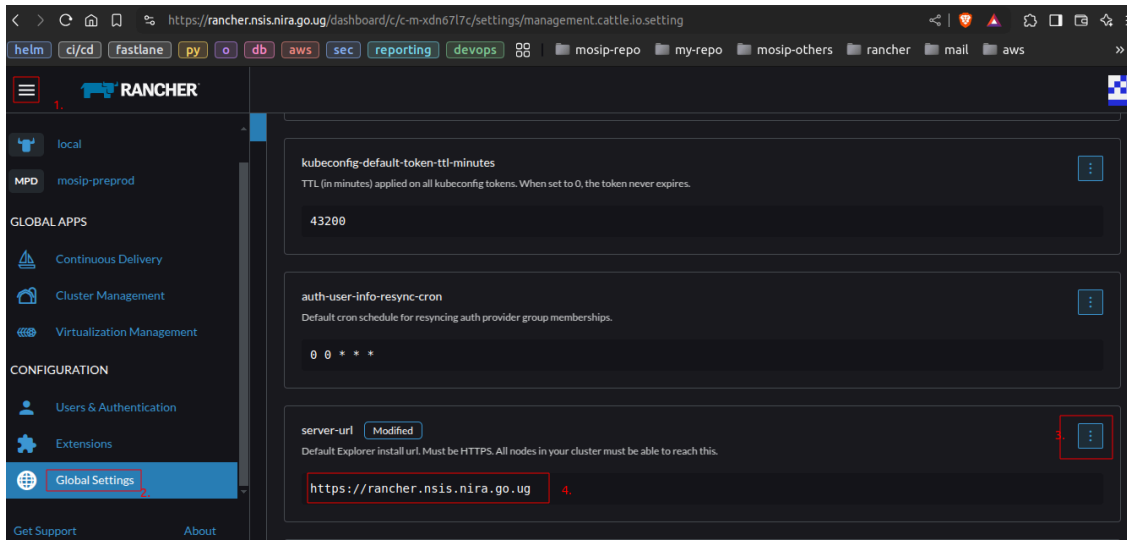
```
helm repo add rancher-latest https://releases.rancher.com/server-charts/latest
helm repo update
helm install rancher rancher-latest/rancher \
  --namespace cattle-system \
  --create-namespace \
  -f rancher-values.yaml \
  --version=2.8.3
```

Login

- Open Rancher page `https://rancher.nsis.nira.go.ug` .
- Get Bootstrap password using

```
kubectl get secret --namespace cattle-system bootstrap-secret -o go-template='{{
.data.bootstrapPassword|base64decode}}{{ "\n" }}
```

- Assign a password. IMPORTANT: makes sure this password is securely saved and retrievable by Admin.
- Remove port 30443 from the server-url section in the Rancher dashboard, if present.



Keycloak deployment

- Navigate to the Keycloak application directory:

```
cd ~/k8s-infra/apps/keycloak/
```

- Ensure that the `tls.crt` and `tls.key` files, which contain the public and private certificates for `*.nsis.nira.go.ug`, are available. Then, create the secret in the Keycloak namespace via the below commands

```
kubectl create ns keycloak
```

```
kubectl create secret tls nira-tls-secret --cert=tls.crt --key=tls.key -n keycloak
```

`tls.crt` : contain the full chain certificate

`tls.key` : contain the private for the certificate

- Update `tls` hosts and secret name containing the SSL certificates `nira-tls-secret` in `values.yaml` file.

```
ingress:
  enabled: true
  ...
  ...
  extraTls:
    - hosts:
      - keycloak.nsis.nira.go.ug
      secretName: nira-tls-secret
```

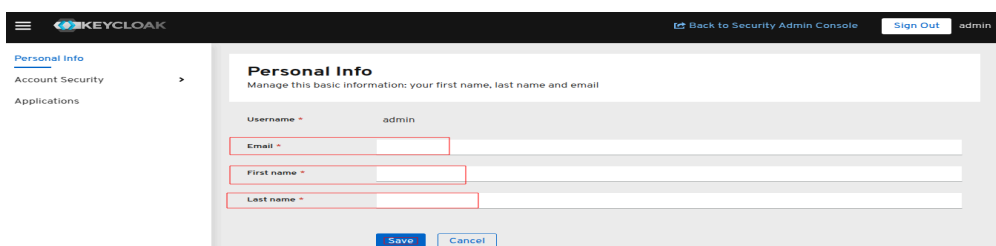
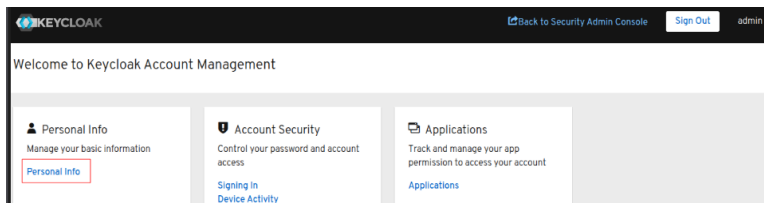
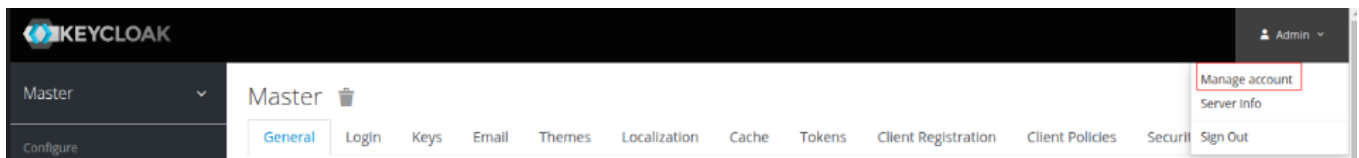
- Pass the path to your kubeconfig file and the Keycloak hostname as parameters.

```
./install.sh <KUBECONFIG-FILE> <keycloak.host.name>
```

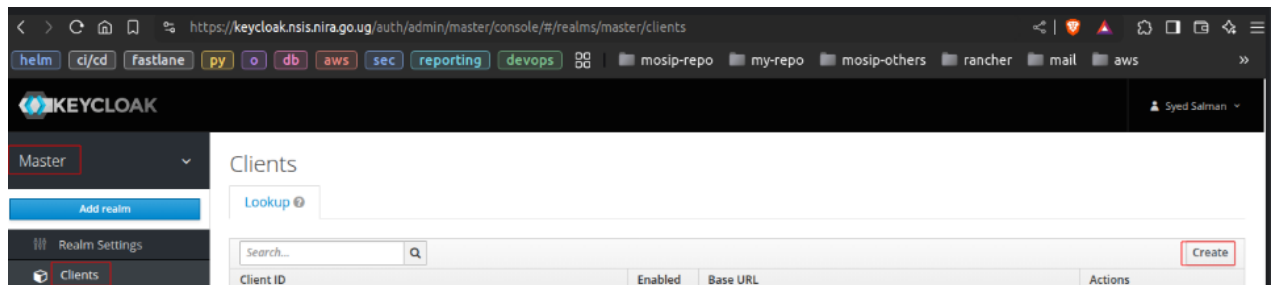
eg. `./install.sh </PATH/TO/KUBECONFIG/FILE> keycloak.nsis.nira.go.ug`

Integrate Keycloak with Rancher UI

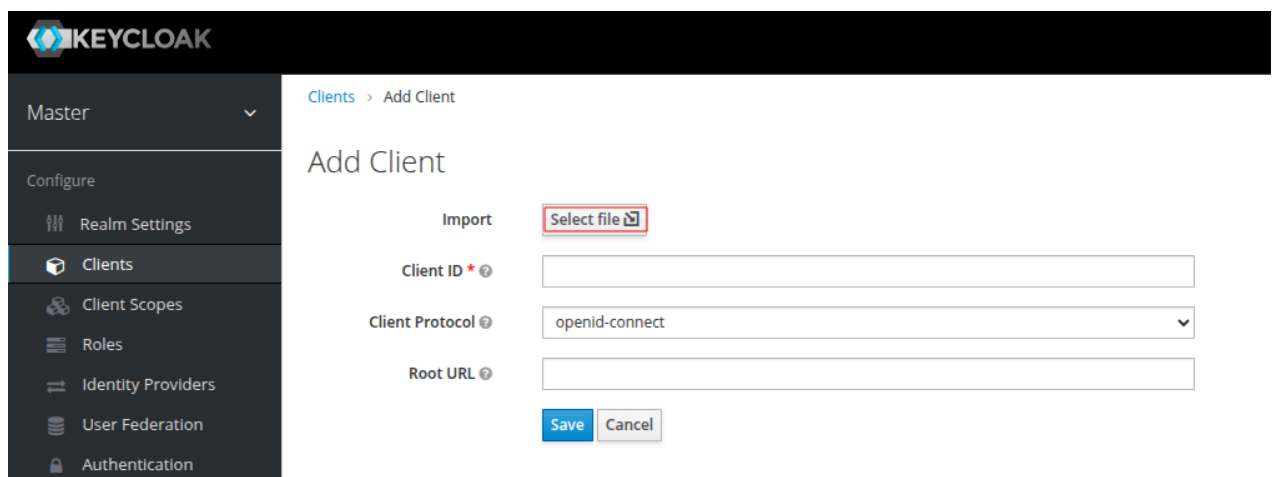
- Log in as the `admin` user in Keycloak and ensure that the `email ID` , `first name` ,and `last name` fields are populated for the admin account. This is essential for Rancher authentication.



- Create client for via `keycloak_client.json` .
 - Navigate to `Keycloak` --> `master realm` --> `Clients` ----> `Create` .



- Click on `Select file` to import client via configuration file `keycloak_client.json` present in keycloak apps directory.



- Update your rancher domain in **Client ID** section, select **Client Protocol** to **saml** ----> click on **save** .

Keycloak Admin Console: Add Client

Client ID:

Client Protocol:

Client SAML Endpoint:

Buttons: **Save** **Cancel**

- Navigate to the newly created saml client, Update domain in **Valid Redirect URIs** section. Ensure to enable the options provided in below images.

Client ID	Enabled	Base URL	Actions
account	True	https://keycloak.nsis.nira.go.ug/auth/realms/master/account/	Edit Export Delete
account-console	True	https://keycloak.nsis.nira.go.ug/auth/realms/master/account/	Edit Export Delete
admin-cli	True	Not defined	Edit Export Delete
broker	True	Not defined	Edit Export Delete
https://rancher.nsis.nira.go.ug/v1-saml/keycloak/saml/metadata	True	Not defined	Edit Export Delete
master-realm	True	Not defined	Edit Export Delete
security-admin-console	True	https://keycloak.nsis.nira.go.ug/auth/admin/master/console/	Edit Export Delete

Keycloak Admin Console: Settings for https://rancher.nsis.nira.go.ug/v1-saml/keycloak/saml/metadata

Client ID: https://rancher.nsis.nira.go.ug/v1-saml/keycloak/saml/metadata

Name: rancher

Description:

Enabled: ☒ ON

Always Display in Console: ☐ OFF

Consent Required: ☐ OFF

Login Theme: keycloak

Client Protocol: saml

Include AuthnStatement: ☐ OFF

Include OneTimeUse: ☐ OFF

https://keycloak.nsis.nira.go.ug/auth/admin/master/console/#/realms/master/clients/68b8ef4c-69ed-4e4e-849c-98380218f274

py o db aws sec reporting devops | mosip-repo my-repo mosip-others

Force Artifact Binding ☐ OFF

Sign Documents ☒ ON

Optimize REDIRECT signing key lookup ☐ OFF

Sign Assertions ☒ ON

Signature Algorithm

SAML Signature Key Name

Canonicalization Method

Encrypt Assertions ☐ OFF

Client Signature Required ☐ OFF

Force POST Binding ☐ OFF

Front Channel Logout ☐ OFF

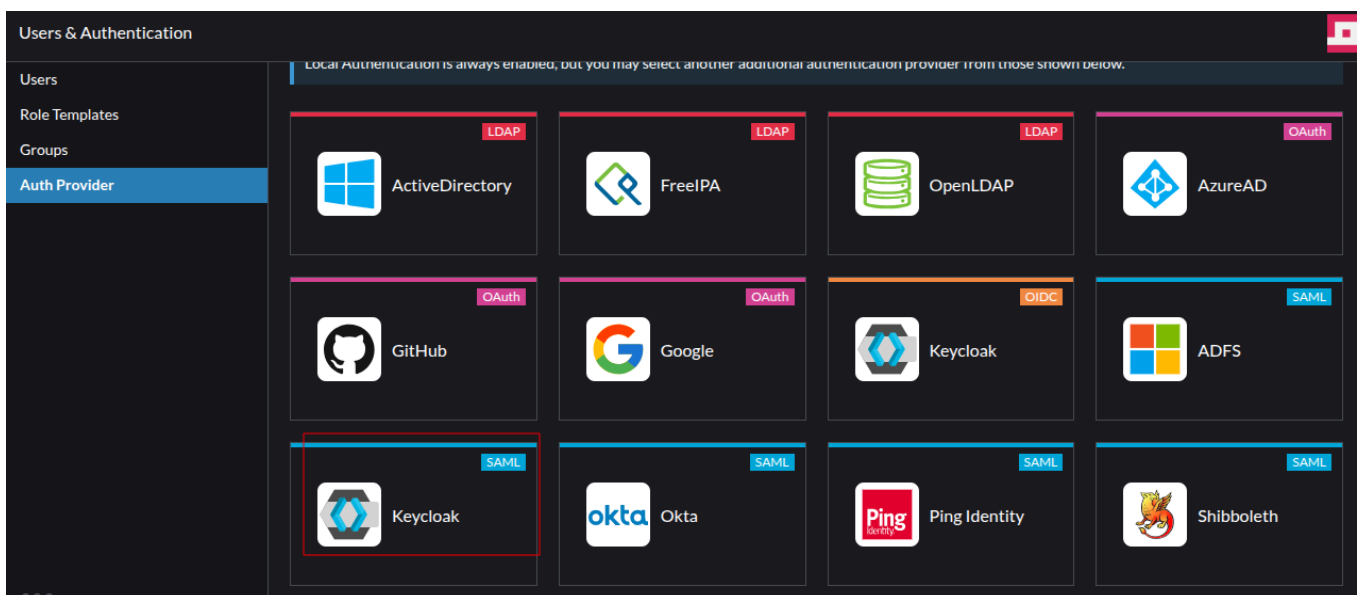
Force Name ID Format ☐ OFF

Name ID Format

Root URL

Valid Redirect URIs

- Navigate to rancher dashboard ----> Users & Authentication ----> Auth Provider ----> Keycloak (SAML) .

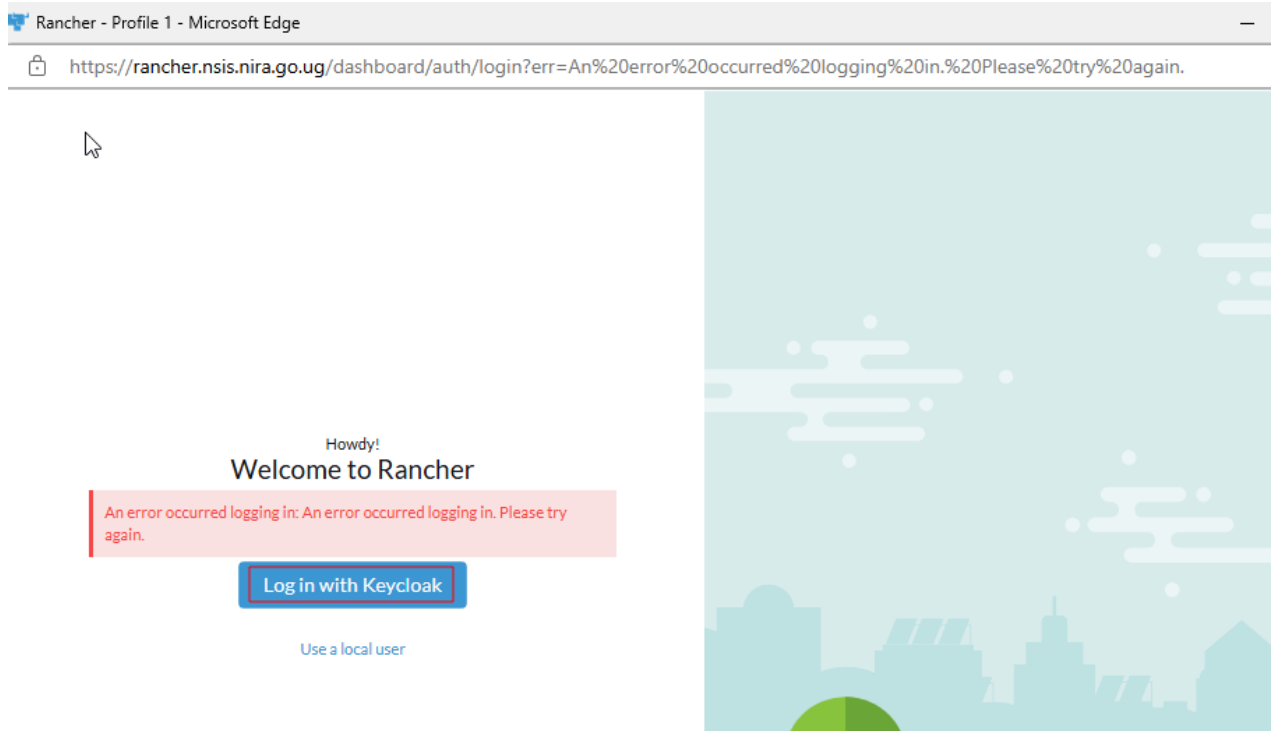


- Specify the following mappings in Rancher's Authentication Keycloak form:
 - Display Name Field: givenName
 - Username Field: email
 - UID Field: username
 - Entity ID Field: <https://your-rancher-domain/v1-saml/keycloak/saml/metadata>
 - Rancher API Host: <https://your-rancher-domain>
 - Groups Field: member
- Copy the content from keycloak url: <https://keycloak-domain/auth/realms/master/protocol/saml/descriptor> and provide it in rancher Auth provider metadata.xml field.
example: <https://keycloak.nsis.nira.go.ug/auth/realms/master/protocol/saml/descriptor>
- For certificate: you can provide valid full chain ssl certificate & private key of *.nsis.nira.go.ug (ensure the certificate to be RSA type).

The screenshot shows the Rancher dashboard's 'Authentication Provider: Keycloak (SAML)' configuration page. The page is titled 'Authentication Provider: Keycloak (SAML) Active'. It contains a 'Configure a Keycloak account' section with the following fields:

- Display Name Field: givenName
- User Name Field: email
- UID Field: username
- Groups Field: member
- Entity ID Field: https://rancher.nsis.nira.go.ug/v1-saml/keycloak/saml/metadata
- Rancher API Host: https://rancher.nsis.nira.go.ug
- Private Key: (A text area containing a long RSA private key, with a 'Read from File' button below it.)
- Metadata XML: (A text area containing SAML metadata XML, with a 'Save' button below it.)

- Click on save , it will pop up new window and click on Login with Keycloak



- Once the login is successful, click on save.

Authentication Provider: Keycloak (SAML) (Active)

The Keycloak authentication provider is currently enabled.

[Edit Config](#)[Disable](#)

Display Name Field: givenName
User Name Field: email
UID Field: username
Entity ID Field: https://rancher.nsis.nira.go.ug/v1-saml/keycloak/saml/metadata
Rancher API Host: https://rancher.nsis.nira.go.ug
Groups Field: member

Configure who should be able to login and use Rancher

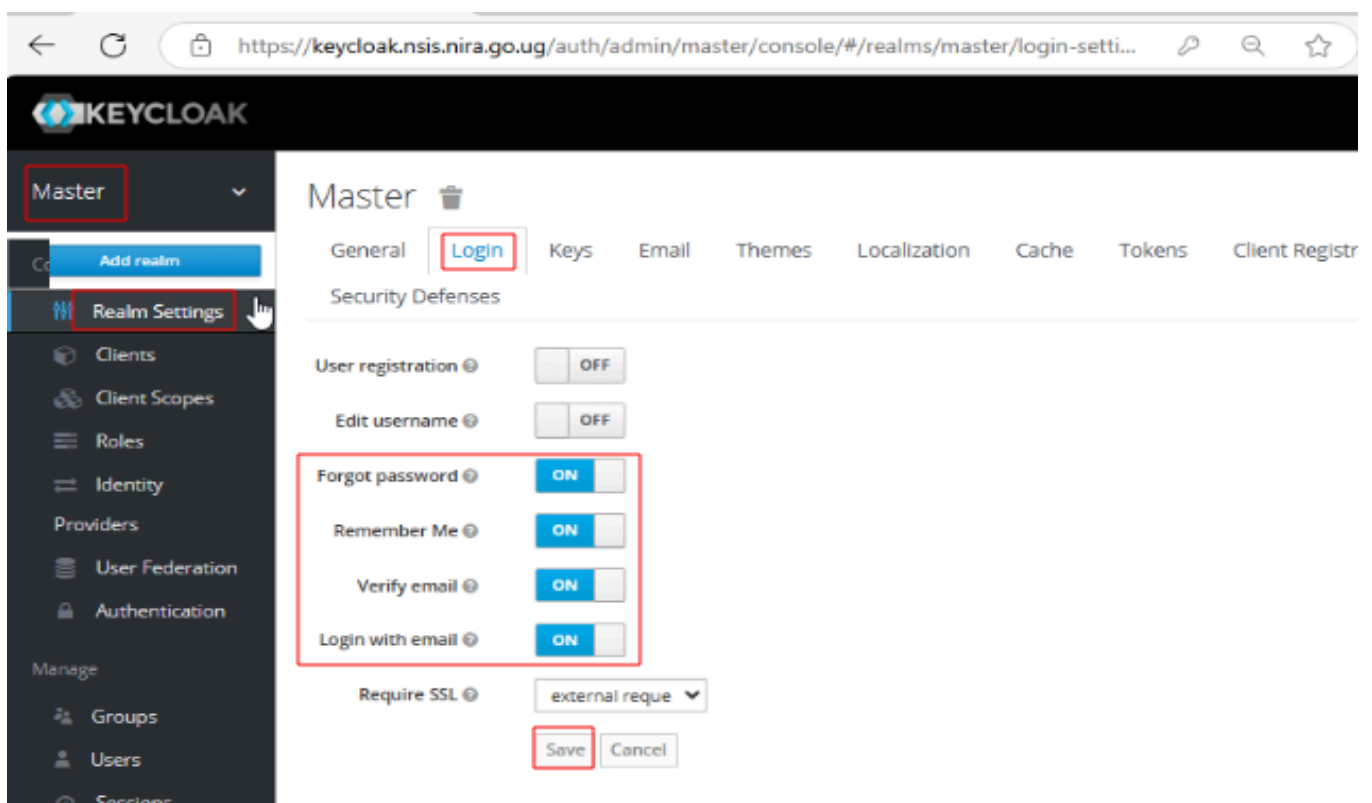
- ☒ Allow any valid user
- ☐ Allow members of clusters and projects, plus authorized users & groups
- ☐ Restrict access to only the authorized users & groups

[Save](#)

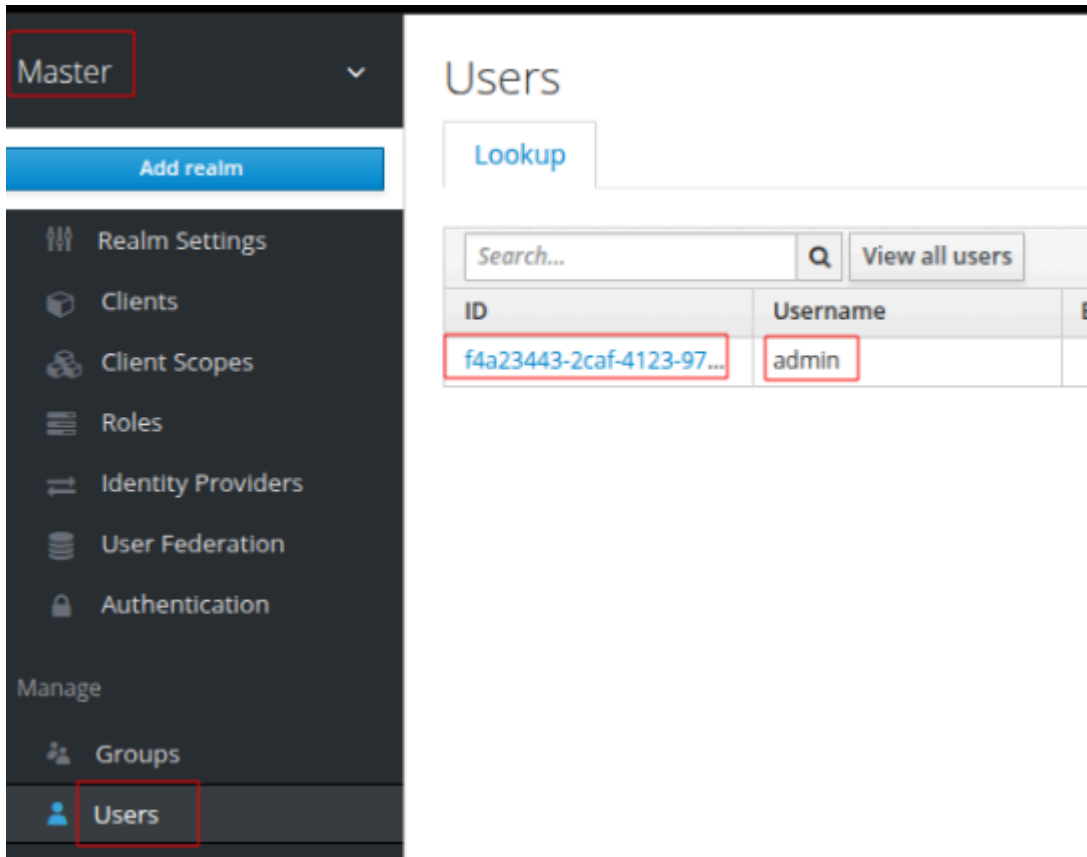
Configuring SMTP and Login Settings in Keycloak

Login Settings

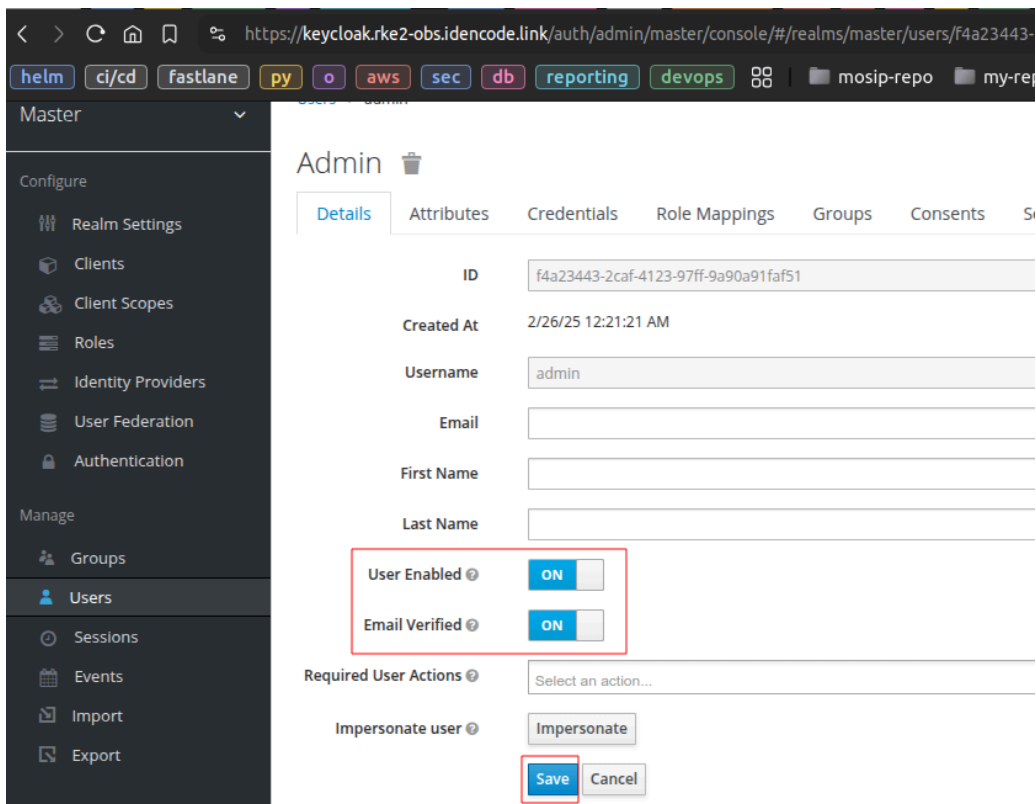
- Navigate to **Master Realm** → **Realm Settings** → **Login** → Enable the options provided in the below image:



- Navigate to **Master Realm** → **Users** → **Search for admin user**.



- Enable **Email Verified** option for admin user.



Configuring SMTP

To enable SMTP and login configurations in Keycloak, follow these steps:

- Before configuring SMTP, ensure that the Keycloak admin user has an email address, first name, and last name specified. Refer to the [integration guide](#) for more details.
- Navigate to **Master Realm** → **Realm Settings** → **Email**.
- Enter the required SMTP details as shown in the image below:

The screenshot shows the Keycloak Admin Console interface for configuring SMTP settings. The browser address bar displays the URL: `https://keycloak.nsis.nira.go.ug/auth/admin/master/console/#/realms/master/smtp-setti...`. The left sidebar is divided into 'Configure' and 'Manage' sections. Under 'Configure', 'Realm Settings' is selected, and under 'Manage', 'Users' is visible. The main content area is titled 'Master' and features a tabbed interface with 'Email' selected. The configuration form includes the following fields and controls:

- Host**: A text input field.
- Port**: A text input field containing the value '587'.
- From Display Name**: A text input field with placeholder text 'Display Name for Sender Email Address'.
- From**: A text input field.
- Reply To Display Name**: A text input field with placeholder text 'Display Name for Reply To Email Address'.
- Reply To**: A text input field with placeholder text 'Reply To Email Address'.
- Envelope From**: A text input field with placeholder text 'Sender Envelope Email Address'.
- Enable SSL**: A toggle switch set to 'OFF'.
- Enable StartTLS**: A toggle switch set to 'ON'.
- Enable Authentication**: A toggle switch set to 'ON'.
- Username**: A text input field.
- Password**: A password input field with masked characters '*****' and a visibility toggle icon.

At the bottom of the form are three buttons: 'Test connection' (blue), 'Save' (red), and 'Cancel' (grey).

- Click **Save** to apply the settings.
- Click **Test Connection** to verify that Keycloak can successfully send emails.

This configuration ensures that Keycloak can send email notifications, such as password resets and account updates, to users.

Monitoring deployment

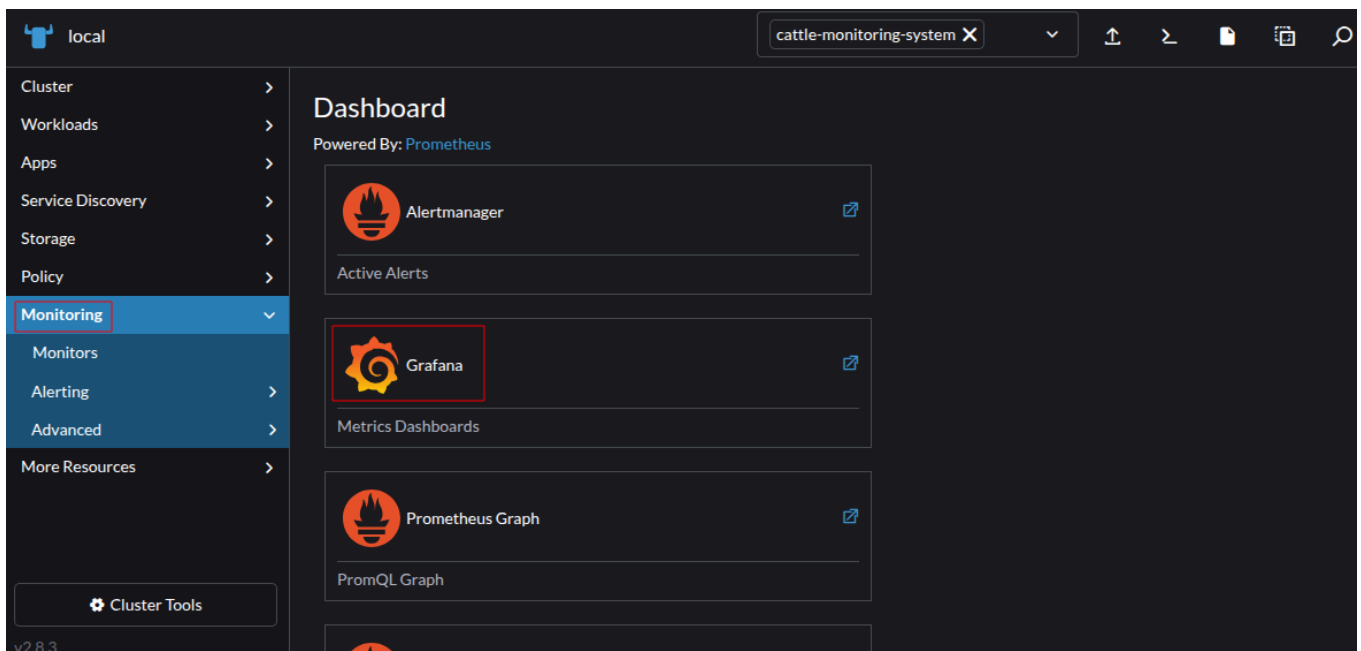
- Navigate to the Monitoring directory:

```
cd ~/k8s-infra/monitoring/
```

- Run `install.sh` to deploy monitoring application. Provide `local` as cluster-id user input variable.

```
./install.sh
....
....
Please enter the env cluster-id: local
```

- To access Grafana dashboard, navigate to the cluster on Rancher Dashboard ---> All namespace ---> Monitoring ---> Grafana Dashboards .



- Use the below command to fetch the `admin` user password for grafana dashboard.

```
kubectl -n cattle-monitoring-system get secret rancher-monitoring-grafana -o json | jq -r  
'data.admin-password' | @base64d'
```

Logging Deployment Guide

Deployment

1. Navigate to the Logging Directory.

Change to the logging directory before proceeding with the deployment:

```
cd ~/k8s-infra/logging/
```

2. Modify the Installation Script.

Before running the installation script, update the following configurations:

- **Comment out** the following lines in `install.sh` to disable Istio add-ons:

```
# echo Install istio addons
# helm -n $NS install istio-addons chart/istio-addons --set kibanaHost=$KIBANA_HOST --set
installName=$KIBANA_NAME
```

- **Update the Kibana Hostname** in `install.sh` :

```
KIBANA_HOST=rancher-kibana.nsis.nira.go.ug
```

3. Deploy the Logging Operator.

Run the `install.sh` script to install the logging operator and associated applications:

```
./install.sh
```

4. Configure Kibana Ingress.

Update the Kibana hostname in `kibana-ingress-ingress.yaml` , then apply the updated configuration:

```
kubectl apply -f kibana-ingress-rancher.yaml
```

5. Configure Elasticsearch Index Lifecycle Policy.

Modify the `./elasticsearch-ilm-script.sh` file as per your requirements.

Then, execute the script to apply the **Index Lifecycle Policy** and **Index Template** to Elasticsearch:

```
./elasticsearch-ilm-script.sh
```

Configure Rancher Fluentd

To collect logs, create **ClusterOutputs** as follows:

- Create an Elasticsearch ClusterOutput:

```
kubectl apply -f clusteroutput-elasticsearch.yaml
```

- Create a ClusterFlow:

```
kubectl apply -f clusterflow-elasticsearch.yaml
```

Dashboards Configuration

- Load Dashboards into Kibana.

Run the following command to import all dashboards from the `./dashboards` directory into Kibana:

```
./load_kibana_dashboards.sh ./dashboards <cluster-kube-config-file>
```